LAB 05

Indicator Stage: Pine Script v6 Indicators and their Python Twins

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Indicators
Duration	25 min
Simulator page	http://localhost:8000/#/data?tab=chart&symbol=SPY&overlays=sma:50,sma:200,bollinger
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's traders read trend (EMA 20/50), volatility (Bollinger Bands) and momentum (RSI) on TradingView, while the pipeline's signals are computed in Python. If the two disagree, a trader and the automated strategy will act on different numbers, so the desk builds Pine Script indicators and exact Python twins and cross-checks them value by value.

Objective

Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 05. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 05: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Overlay trend indicators

Data > Chart for SPY: toggle sma:50, sma:200, ema:20 and bollinger.

Open: http://localhost:8000/#/data?tab=chart&symbol=SPY&overlays=sma:50,sma:200,bollinger (Data > Chart SPY with SMA 50/200 and Bollinger)

Check: Lines draw over the candles (TradingView Lightweight Charts).

Step 2. Oscillators

Switch the lower pane between RSI(14) and MACD.

Open: http://localhost:8000/#/data?tab=chart&symbol=SPY&osc=macd (Data > Chart SPY MACD)

Check: RSI has 30/70 guide lines; the MACD histogram changes colour above and below 0.

Step 3. Export Pine Script v6

Strategy Library > EMA crossover > Pine Script v6 > Copy. In TradingView open the Pine Editor, paste, 'Add to chart'.

Open: http://localhost:8000/#/strategies?key=ema_cross (Strategies > EMA crossover)

Check: TradingView shows the two EMAs; their last values match the app's chart.

Expected in the app: A chart with overlays and an oscillator pane, and a Pine Script pasted into TradingView whose values match.

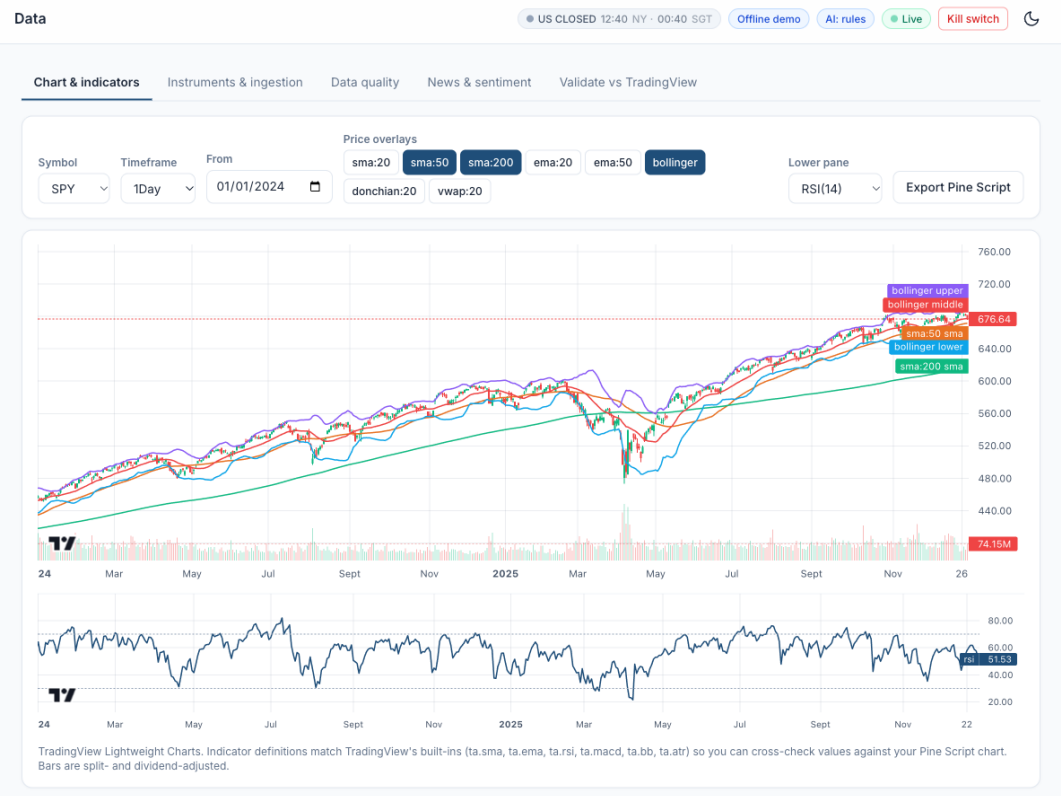


Figure: Lab 05 in the Trading Simulator.

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Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Indicators and their Pine Script twin - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: for SPY 2018-2025 compute SMA(50), SMA(200), EMA(20), Bollinger Bands (20, 2), RSI(14) and MACD (12, 26, 9) with pandas only, print the latest values, and plot the price with the overlays and RSI and MACD in panes below. Then write the same indicators as a TradingView Pine Script v6 indicator I can paste into the Pine Editor.

Open in Claude | **Open in ChatGPT** | **Open Google Colab**

Debrief

Technical indicators turn prices into trend, volatility and momentum readings; defining them once in Pine Script and once in Python, and proving they match, keeps traders and the automated pipeline on the same numbers. Faster settings fire more signals, which means more trades and more cost.

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